Uber Ride Analytics | Power BI Dashboard

By: **Rishan Batra** · Tool: Power BI Desktop · Data: Uber ride-booking dataset (150K rows)

Power BI DAX Power Query

Project Overview

Ride-hailing operations generate three kinds of pain for the teams that run them: **bookings that never convert into revenue**, **fleets that perform unevenly across vehicle types**, and **demand that clusters unpredictably by time and location**. Raw trip logs don't answer *which vehicle type is underperforming*, *where money is lost to cancellations*, or *when to surge drivers* - someone has to model and visualize the data first.

This project turns a raw 150,000-row Uber ride log into a **decision-ready Power BI report**: sourcing messy transactional data, modeling it in Power Query, writing DAX measures for the metrics that matter, and designing an interactive, five-page report rather than a single static chart-dump.

Business questions the dashboard answers: How many bookings convert to completed rides vs. get lost to cancellations or nodriver-found? Which vehicle type drives the most revenue, and which is weakest? Where is demand concentrated by zone and hour? Who are the highest-value customers? Why are customers cancelling, and can it be reduced operationally?

Uber Problems and Business Requirements

The report was scoped page-by-page against a fixed set of business requirements, each mapped to a specific DAX filter or function.

1. Overview Page - KPIs. Completed Bookings (dax filter = "Completed"); Lost Bookings (dax filter <> "Completed");
Revenue (sum of booking value); Total Distance (sum of distance); Avg. Distance (average DAX function). *Vehicle filter applied across all downstream pages.* Also: Bookings Completed per month/quarter (calendar function); Revenue per month/quarter; Revenue by Vehicle Type; Top Drop & Pickup Location by booking count (Top-N filter, N=1); Avg. customer's and driver's rating (average DAX function).

2. **VehiclePage-detailbyvehicle.** Vehicle images; distinct Customers (distinct count); Revenue; Completed Bookings; Contribution % (divide, with filters removed via ALL); Total Distance per month as a sparkline.
3. **Revenue Page.** Revenue per day (date function); Revenue by payment method; Top 5 Drop Locations by revenue (Top-N, N=5); Top 5 Customers by revenue (distinct count of customers).
4. **Bookings Page.** Ride distance per month/quarter; Reason for cancellation by customers; Bookings by pickup location; Bookings per week by time slot (DAX switch/return function) with an added **Target Bookings KPI** gauge.

Key Insights & KPIs

Metric	Value
Total Bookings	150,000
Completed Bookings	93,000 (62%)
Lost Bookings (cancelled / no driver / incomplete)	57,000 (38%)
Total Revenue	₹47.26M – 51.85M*
Total Distance Travelled	2.51M km
Avg. Distance per Ride	24.6 km
Avg. Customer Rating	4.40 / 5
Avg. Driver Rating	4.23 / 5

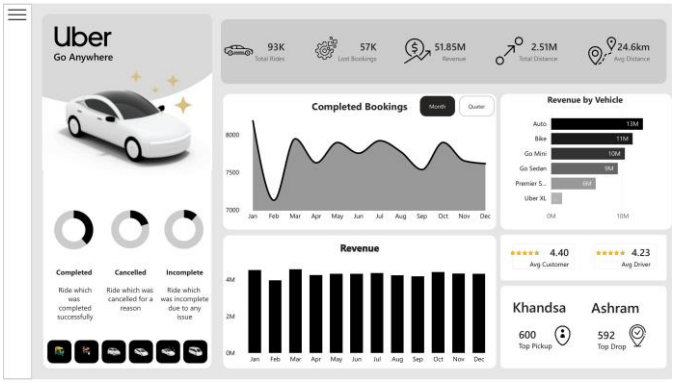
*Revenue is reported at two scopes: ₹47.26M reflects completed-ride booking value only; ₹51.85M on the KPI strip includes incomplete trips with a partial fare captured. Both are intentional — an "operations" view vs. a "gross" view.

What the data shows:

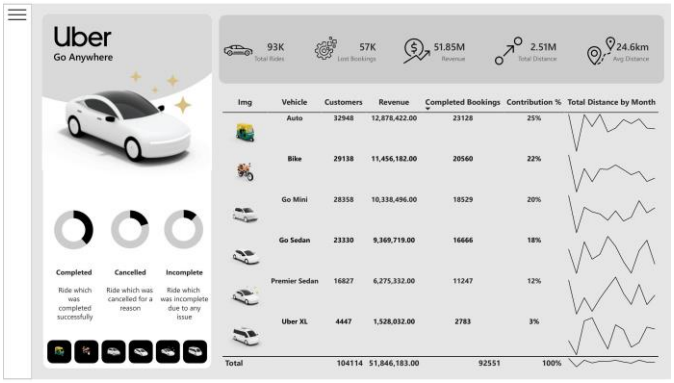
- **Auto and Bike lead the fleet mix** — together 47% of completed bookings and revenue, while **Uber XL is the long tail** at just 3% of contribution despite the highest per-ride value.
- **UPI dominates payments** at 45% of revenue, more than 3× Cash — a signal for where to invest in payment-partner reliability.

- **6 PM–9 PM is the busiest slot** every day of the week (33K bookings), the clearest signal for driver-supply incentives.
- **Khandsa and Ashram are the top pickup/drop nodes**, with a fairly even spread across the next five zones — demand is distributed, not concentrated.
- **"Wrong Address" and "Change of Plans"** are the top two customer-cancellation reasons, both addressable via UX rather than driver-side fixes.
- Revenue per day is volatile but range-stable (₹0.1M–0.3M), with no strong seasonal trend — growth looks volume-driven, not seasonal.

Dashboard Visualizations



1 · Overview— top-line KPIs, trends, vehicle revenue, ratings, ride-outcome rings.



2 · Vehicle — per-vehicle customers, revenue, contribution %, distance sparkline.



3 · Revenue — daily revenue, payment mode, top drop locations & customers.



4 · Bookings — ride distance, cancellation reasons, pickup locations, time-slot matrix, Target KPI gauge.

Tech Stack

- **Power BI Desktop** — report authoring, data modeling, page navigation
- **Power Query (M)** — data cleaning, column typing, calendar table generation, unpivoting time-slot data
- **DAX** — all KPIs, ratios, and time-intelligence measures
- **Datasource** — Uber ride-booking transactional log (150,000 rows: booking status, vehicle type, fare, distance, ratings, payment method, pickup/drop location, timestamps) **Core DAX measures used across the report:**

```
Completed Bookings =
CALCULATE(
   COUNTROWS(Bookings),
   Bookings[Booking Status]="Completed" )
```

```
)
```

```
Revenue =
SUM(Bookings[Booking Value])
```

```
Lost Bookings =
CALCULATE(
   COUNTROWS(Bookings),
   Bookings[Booking Status]<>"Completed"
```

```
Total Distance =
SUM(Bookings[Ride Distance])
```

```
Avg Distance =
AVERAGE(Bookings[Ride Distance])
```

```
   [Revenue],
   CALCULATE([Revenue],
      ALL(Bookings[Vehicle Type])) )
```

```
Distinct Customers =
DISTINCTCOUNT(Bookings[Customer ID])
```

```
Target Bookings KPI =
VAR Target = 149000
RETURN DIVIDE(
   [Completed Bookings], Target)
```

```
Contribution % = DIVIDE(
```

How to Interact

- **Page navigation** - the pill-style nav bar (Home · Overview · Vehicle · Revenue · Bookings) on the landing page and the left sidebar icons on every report page switch between the five views.
- **Vehicle filter** - present on the Overview and Vehicle pages; slices every KPI, chart, and table by a single vehicle type (Auto, Bike, Go Mini, Go Sedan, Premier Sedan, Uber XL) or shows them unfiltered.
- **Month / Quarter toggle** - on the Overview (Completed Bookings) and Bookings (Ride Distance) pages, switches time granularity of the trend line without changing the visual.
- **Date-range slider** - on the Revenue page, drag either handle under ”Revenue per Day” to scope the daily revenue trend and the two Top-5 tables to a custom window.
- **Drill-down tables** - the Vehicle page table and the Bookings-per-Week-by-Time-Slot matrix support Power BI’s native drilldown/expand to inspect individual cells.
- **Cross-filtering** - clicking any bar, donut segment, or table row (e.g., a vehicle type, a payment method, a cancellation reason) cross-filters every other visual on that page, standard Power BI report interaction behavior.